

Filipino Adaptation of OmniVoice for Voice Cloning and Text-to-Speech

Final Project Presentation

Current Trends

June 11, 2026

The Problem

Voice cloning systems synthesize speech in a target speaker's voice from a short reference audio.

- High-quality voice cloning is strongest in high-resource languages.
- Filipino speech AI remains low-resource, with few open Filipino workflows.
- OmniVoice supports Filipino, but with only about **7.71 hours** of Filipino in its training mixture.
- The UP-DSP Philippine Languages Database holds almost **49 hours** of Filipino speech.

What We Did

Fine-tune OmniVoice on curated Filipino speech and measure whether the adapted model is better than the base model.

Background: OmniVoice

OmniVoice: Towards Omnilingual Zero-Shot Text-to-Speech with Diffusion Language Models

Hao Zhu, Lingmin Ye, Wei Kang, Zengwei Yan, Liyong Cao, Fangjun Kong,
Zhifeng Han, Meng Zhang, Long Lin, Daniel Povey
Kuaishou, China
(zhaokan3,dpovey@kuaishou.com)

Abstract

We present OmniVoice, a massively multilingual zero-shot text-to-speech (TTS) model that scales to over 600 languages. At its core is a novel diffusion language model-style discrete-continuous autoregressive (NAR) architecture. Unlike conventional discrete NAR models that suffer from performance bottlenecks in complex two-stage (text-to-semantic-to-acoustic) pipelines, OmniVoice directly maps text to multi-codebook acoustic tokens. This simplified approach is facilitated by two key technical innovations: (1) a full-codebook random masking strategy for efficient training, and (2) initialization from a pre-trained LLM to ensure superior intelligibility. By leveraging a 50k-hour multilingual dataset curated entirely from open-source data, OmniVoice achieves the broadest language coverage to date and delivers state-of-the-art performance across Chinese, English, and diverse multilingual benchmarks. The code and pre-trained models are publicly available.

1 Introduction

Zero-shot text-to-speech (TTS) models trained on large-scale datasets have demonstrated a remarkable ability to generate high-quality speech conditioned on only a few seconds of reference audio [1, 2]. Despite these advances, most existing models support only a limited set of languages, often leaving hundreds of low-resource languages behind. Expanding language coverage is not merely a technical challenge but also a critical step toward extending speech technologies to languages across the globe. To address this gap, we aim to develop a massively multilingual zero-shot TTS model supporting hundreds of languages. Realizing such extensive scaling requires a TTS architecture with exceptional capacity and robustness to handle diverse linguistic patterns.

In the pursuit of such scalable and high-quality TTS, current research primarily adheres to two paradigms: autoregressive (AR) [3, 4, 5, 6, 7, 8, 9, 10, 11] and non-autoregressive (NAR) models [12, 13, 14]. NAR models offer advantages in both inference speed (via parallel decoding) and robustness (owing to bidirectional context) [12, 14, 15]. Unlike AR frameworks, models can be broadly categorized into continuous-latent based [14, 16, 17, 18] and discrete-token based [19]. The latter was shown to yield superior prosodic diversity [20].

However, state-of-the-art (SOTA) discrete-token NAR systems [19] typically rely on complex two-stage cascaded pipelines (text-to-semantic followed by semantic-to-acoustic). While such decoupling simplifies the training of individual modules, it also introduces significant drawbacks: (1) error propagation, where inaccuracies in semantic prediction degrade the final audio quality; and (2) information bottlenecks, where low-bitrate semantic representations sacrifice fine-grained acoustic details. While single-stage alternatives [21] attempt to bypass these issues, they have historically lagged behind two-stage systems in terms of speech intelligibility.

¹<https://github.com/K2-fsa/omnivoice>

Preprint. Under review.

- Open-source multilingual zero-shot TTS and voice cloning model.
- Supports 600+ languages, including Filipino (fil).
- Generates discrete acoustic tokens with a masked-token (diffusion language model) process.
- Provides official fine-tuning and evaluation scripts.
- Practical to adapt instead of training a Filipino model from scratch.

Source: Zhu et al. (2026).

Research Question

Main Research Question

How much does Filipino-specific fine-tuning improve OmniVoice for Filipino TTS and voice cloning across intelligibility, naturalness, and speaker similarity metrics?

- Model: k2-fsa/OmniVoice, adapted with supervised fine-tuning.
- Data: Filipino read speech from the UP-DSP Philippine Languages Database.
- Compute: free Kaggle T4 GPU notebooks.
- Baseline: the unmodified pretrained OmniVoice checkpoint.
- Tracking: all runs logged in Weights and Biases.

Dataset: UP-DSP Philippine Languages Database



The screenshot shows the Mozilla Data Collective interface for the 'UP - DSP - Philippine Languages Database (UP-DSP-PLD)'. The page includes a header with navigation links: 'Browse Datasets', 'API', 'Contribute Datasets', and 'Data Assistant'. The main title is 'UP - DSP - Philippine Languages Database (UP-DSP-PLD)'. Below the title, it specifies the license as 'CC-BY-NC-4.0' and the steward as 'UP EEEI - Digital Signal...'. Metadata includes 'Task: ASR', 'Release Date: 3/19/2026', 'Format: WAV, LOG', and 'Size: 45.63 GB'. There are buttons for 'Download' and 'Connect API', along with a 'Share' icon. A 'Files' section shows 'Latest Version'. A 'Description' section states: 'This dataset contains multilingual, text and speech pairs for ten Philippine languages namely Filipino, English, ...'.

- Released through Mozilla Data Collective (CC-BY-NC-4.0).
- WAV audio with LOG transcript files.
- Filipino subset: 135 speakers, 52,879 utterances, 48:56:36.
- We used read speech only: the transcript reliably matches the audio.

Source: Cajote et al. (2024).

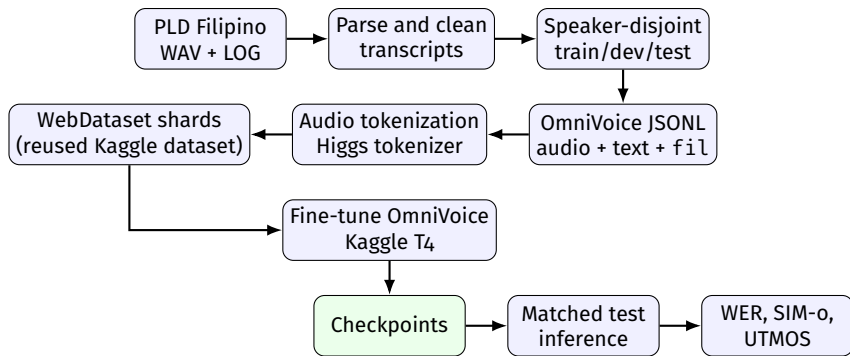
Data Preparation

Conservative filtering: read speech only, no transcripts with digits or parentheses, no broken WAVs, clips between 1 and 15 seconds, speaker-disjoint splits.

Split	Samples	Percent	Hours	Speakers
Train	33,448	79.1%	31.9	103
Development	4,506	10.7%	3.7	13
Test	4,322	10.2%	4.2	13
Total	42,276	100%	39.8	129

The 31.9 training hours are about $4\times$ the Filipino hours in OmniVoice's original training mixture. Domains include news, medical sentences, stories, greetings, proverbs, and many isolated-word prompts.

Pipeline



Audio is tokenized once into 8-codebook acoustic tokens and reused across runs; training recovers masked acoustic tokens from text, language, and prompt audio.

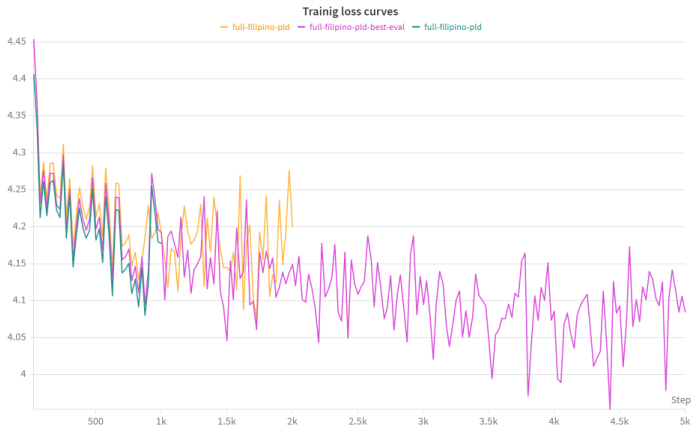
Fine-Tuning Setup

- Environment: Kaggle notebooks, $2\times$ NVIDIA T4, FP16, SDPA attention, gradient accumulation.
- Start from the pretrained k2-fsa/OmniVoice checkpoint.
- Loss: OmniVoice's own objective, weighted cross-entropy on masked acoustic tokens.
- Optimizer: AdamW, cosine learning-rate decay, 1% warmup, seed 42.

Run	Learning rate	Steps	Checkpoint policy
FT-5e-6	5e-6	5000	lowest dev loss
FT-1e-5	1e-5	5000	lowest dev loss (found at step 4900)
FT-2e-5	2e-5	5000	lowest dev loss

A **controlled comparison**: same steps, same checkpoint policy, only the learning rate differs. (Two earlier exploratory runs at 1000 and 2000 steps with final-step checkpoints are project history.)

Training Behavior



W&B training-loss curves for the fine-tuning runs, including the preliminary exploratory runs.

How We Evaluated

All systems generate the same 4,322 held-out test utterances, cloning a reference voice from the same speaker (a different utterance, so the model never hears the target audio).

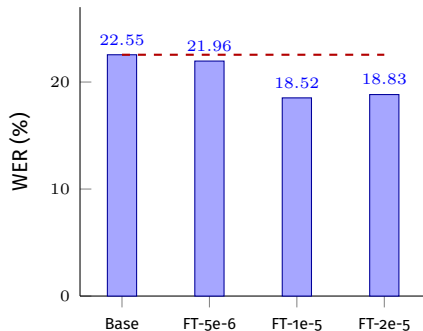
- **WER: “Did it say the right words?”**
Whisper-large-v3 transcribes the generated audio; transcript vs. target text. Lower is better.
- **SIM-o: “Does it sound like the reference speaker?”**
Cosine similarity of speaker embeddings (WavLM ECAPA-TDNN). Higher is better.
- **UTMOS: “Does it sound natural?”**
A learned mean-opinion-score predictor. Higher is better.

Same test split, prompts, inference settings, and evaluator models for every system, so differences come from the checkpoints.

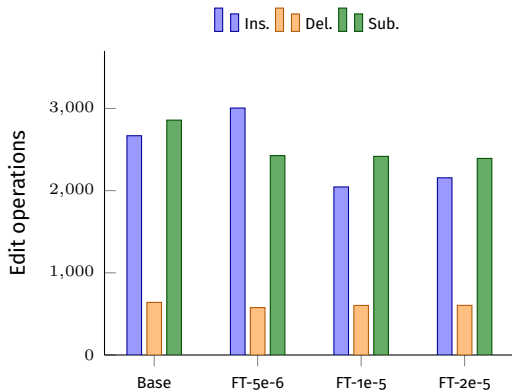
Results: Main Comparison

Model	WER (%)	SIM-o	UTMOS
Base	22.55	0.602	3.64
FT-5e-6	21.96	0.605	3.60
FT-1e-5	18.52	0.604	3.61
FT-2e-5	18.83	0.583	3.61

FT-1e-5 cuts WER by **4.03 points** (17.9% relative) while keeping SIM-o above base. FT-2e-5 is close in WER but loses speaker similarity.



Results: Where the Improvement Comes From



- FT-1e-5 wins by producing **fewer extra words** (insertions: 2,668 $\rightarrow$ 2,045) and **fewer wrong words** (substitutions: 2,858 $\rightarrow$ 2,417).
- FT-2e-5 shows the same pattern, slightly behind on insertions.
- FT-5e-6 cut substitutions but its insertions rose to 3,005, absorbing most of the gain.
- A run can improve one error type while worsening another.

Discussion

- The learning rate controls a **tradeoff**: $1e-5$ and $2e-5$ buy intelligibility; $5e-6$ stays closest to the base model's speaker similarity.
- FT- $2e-5$ shows the cost of pushing harder: nearly the same WER as FT- $1e-5$, but the **only SIM-o below base** (0.583).
- The base model keeps the highest UTMOS (3.64); fine-tunes are close at 3.60–3.61 — a small, consistent naturalness cost.
- No single metric picks the winner; FT- $1e-5$ wins on the **overall pattern**: nearly all the WER gain with SIM-o preserved.
- Practical lessons: tokenize once and reuse; a conservative FP16 + gradient-accumulation recipe makes T4 training feasible.

Limitations

- Automatic metrics only: WER is ASR-mediated (Whisper), SIM-o depends on the speaker encoder, UTMOS is a predictor, not a listening survey.
- Read speech only, with many short isolated-word prompts; says little about spontaneous speech or long-form prosody.
- Filipino only; other Philippine languages in PLD were not tested.
- Three learning-rate settings, not a full sweep, because of free-GPU compute limits.

Conclusion and Recommendations

Answer to the Research Question

Filipino fine-tuning improved OmniVoice intelligibility: WER dropped from 22.55% to 18.52% (17.9% relative) with speaker similarity preserved. The learning rate controls an intelligibility-versus-speaker-similarity tradeoff; no setting wins every metric.

- Recommended checkpoint: **FT-1e-5** (best dev loss, found at step 4900).
- Next: Filipino listener study to validate the automatic metrics.
- Next: more learning rates around 1e-5; repeat runs across seeds.
- Next: curated sentence-level data to improve prosody.

Application Use Case: Voice Cloning

VOICE CLONE

Upload a short clip with its transcript and save a reusable voice for later generation.

- ☐ Name your voice, upload reference audio, and add the matching transcript.
- ☐ Keep cloned voices organized in one library.
- ☐ Reuse the same saved voice across future generations.

OPEN CLONE →

UPLOAD

RECORD

RECORD REFERENCE AUDIO

0:12

Aim for 10-20 seconds. Recording stops automatically at 60 seconds.

RECORDING...

STOP

CLEAR

VOICE NAME

Aria (warm, close-mic)

TRANSCRIPT

The quick brown fox jumps over the lazy dog near the riverbank.

LANGUAGE

English

TRY EXAMPLE VOICE

CLONE VOICE

References

- Cajote, R. D., Guevara, R. C. L., Bayona, M. G. A. R., & Lucas, C. R. G. (2024). *Philippine Languages Database: A multilingual speech corpora for developing systems for Philippine spoken languages*. SIGUL 2024. <https://aclanthology.org/2024.sigul-1.32/>
- Le, M., Vyas, A., Shi, B., Karrer, B., Sari, L., Moritz, R., Williamson, M., Manohar, V., Adi, Y., Mahadeokar, J., et al. (2023). *Voicebox: Text-guided multilingual universal speech generation at scale*. NeurIPS, 36, 14005–14034.
- Saeki, T., Xin, D., Nakata, W., Koriyama, T., Takamichi, S., & Saruwatari, H. (2022). *UTMOS: UTokyo-SaruLab system for VoiceMOS Challenge 2022*. Interspeech 2022. <https://arxiv.org/abs/2204.02152>
- UP EEEL - Digital Signal Processing Laboratory. (2026). *UP-DSP Philippine Languages Database*. Mozilla Data Collective. <https://mozilladatacollective.com/datasets/cmmxhw46c00tqnw07xyr94zjk>
- Wang, C., Chen, S., Wu, Y., Zhang, Z., Zhou, L., Liu, S., et al. (2023). *Neural codec language models are zero-shot text to speech synthesizers*. arXiv. <https://arxiv.org/abs/2301.02111>
- Zhu, H., Ye, L., Kang, W., Yao, Z., Guo, L., Kuang, F., Han, Z., Zhuang, W., Lin, L., & Povey, D. (2026). *OmniVoice: Towards omnilingual zero-shot text-to-speech with diffusion language models*. arXiv. <https://arxiv.org/abs/2604.00688>
- Weights & Biases experiment report:
<https://wandb.ai/duncanb013-polytechnic-university-of-the-philippines/OmniVoice-PLD-Filipino>